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Relationship between northwestern Tasmania and East Gondwanaland in the Late Cambrian/Early Ordovician: Paleomagnetic evidence

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Abstract. The 71 samples from 11 sites in the lower Upper Cambrian turbiditic red siltstone and mudstone of northwestern Tasmania have revealed a high-temperature component which passes a fold test, indicating an age older than Late Devonian. The mean direction of $D = 58.3°$, $I = 11.1°$, $\alpha_{95} = 10.4°$, $k = 20.2$, is interpreted as of primary or early diagenetic origin, and gives a paleomagnetic pole position at (19.4°S, 28.9°E) with $DP = 5.3°$, $DM = 10.5°$. This pole position falls on the Late Cambrian to Early Ordovician part of the East Gondwanaland apparent polar wander path, indicating that if northwestern Tasmania were separated from East Gondwanaland during the Neoproterozoic breakup of the supercontinent Rodinia, it had been accreted to East Gondwanaland by at least the Early Ordovician. Two overprint poles, one at (50.4°S, 177.4°E) with $DP = 6.5°$, $DM = 6.9°$, and the other at (62.0°S, 102.6°E) with $DP = 9.9°$, $DM = 11.3°$, suggest that there could have been more than one overprinting event in the region during the breakup between Australia and Antarctica in the late Mesozoic and early Tertiary.

Introduction

Tasmania, an island some 250 km south of mainland Australia, has the only known Precambrian rocks in the Phanerozoic Tasman Fold Belt (Figure 1). The western three-quarters of the island (Western Tasmania terrane, *Banks et al.* [1989]) was deformed and metamorphosed to various degrees during the 780 to 725 Ma Penguin Orogeny [*Turner et al.*, 1989] although rocks in the Rocky Cape block in the northwestern corner of the island largely escaped this deformation and metamorphism, which was most intense in the Tyennan region of central Tasmania (Figure 1). The older Precambrian rocks in the Rocky Cape block are overlain above a low-angle regional unconformity by a succession of little-metamorphosed, uppermost Precambrian to Cambrian sedimentary and volcanic rocks [*Brown et al.*, 1989].

Numerous hypotheses exist on the tectonic relationship of west Tasmania with respect to mainland Australia. Since no Precambrian rocks had been found in outcrop in the Tasman Fold Belt north of Tasmania (Figure 1), *Veevers and Powell* [1984] regarded west Tasmania as a promontory of Precambrian Australia into the Tasman Fold Belt. More recent studies on the connection between North Victoria Land in Antarctica and the Adelaide–Kanmantoo region of South Australia [e.g., *Borg and DePaolo*, 1991; Stump, 1987, 1992; *Findlay*, 1987, 1991; *Flöttman et al.*, 1993] highlighted the position of Tasmania outboard of the cratonic margin of Precambrian Australia–Antarctica, from which it is separated by a zone of turbidites (the Kanmantoo and Wilson terranes) deformed, metamorphosed and intruded by I-type granites in the Early Cambrian to Early Ordovician (summarized in *Stump* [1995]). This information, together with the difficulty of matching the mid-Neoproterozoic Penguin deformation in Tasmania with any known fold belt in correlative parts of eastern Australia, led *Powell* [1992] to speculate that Tasmania could have been allochthonous to the Tasman Fold Belt in the Neoproterozoic, and welded to the Tasman Fold Belt during the Late Cambrian. *Gulson et al.* [1991] showed that the lead isotope signature of the Cambrian Mount Read Volcanic Belt in west Tasmania (Figure 1) is completely different from those of the Cambrian massive sulphide occurrences in the Kanmantoo and other parts of mainland southeast Australia, and *Elliot and Gray* [1992] suggested that Tasmania lay east of the Tasman Fold Belt during the Paleozoic, and was not emplaced in its Gondwanaland position until the Mesozoic. More recently, west Tasmania has been interpreted as a continental fragment which was part of the rift between Laurentia and Australia/Antarctica in Rodinia in the early Neoproterozoic, and then possibly separated from Australia/Antarctica during the ~700 Ma Rodinian breakup, before accreting to East Gondwanaland during the mid–Cambrian to Early Ordovician Ross–Delamerian Orogeny [*Powell et al.*, 1994].

Being a continental fragment possibly derived from within Rodinia [e.g., *Dalziel*, 1992a, b], and being the only region in the Tasman Fold Belt where Precambrian outcrops have been identified, west Tasmania may thus hold the key for understanding the breakup of Rodinia, and the nature of the Tasman Fold Belt. It is our intention to investigate systematically the Neoproterozoic–early Paleozoic paleomagnetic record of west Tasmania. In this paper we present results from Upper Cambrian turbidite successions in the Rocky Cape region, far northwestern Tasmania (Figures 1 and 2), and interpret their tectonic significance.

Geology of the Study Region

The studied section occurs in the upper part of the Smithton Basin, a gently folded 4-km-thick mixed succession of uppermost

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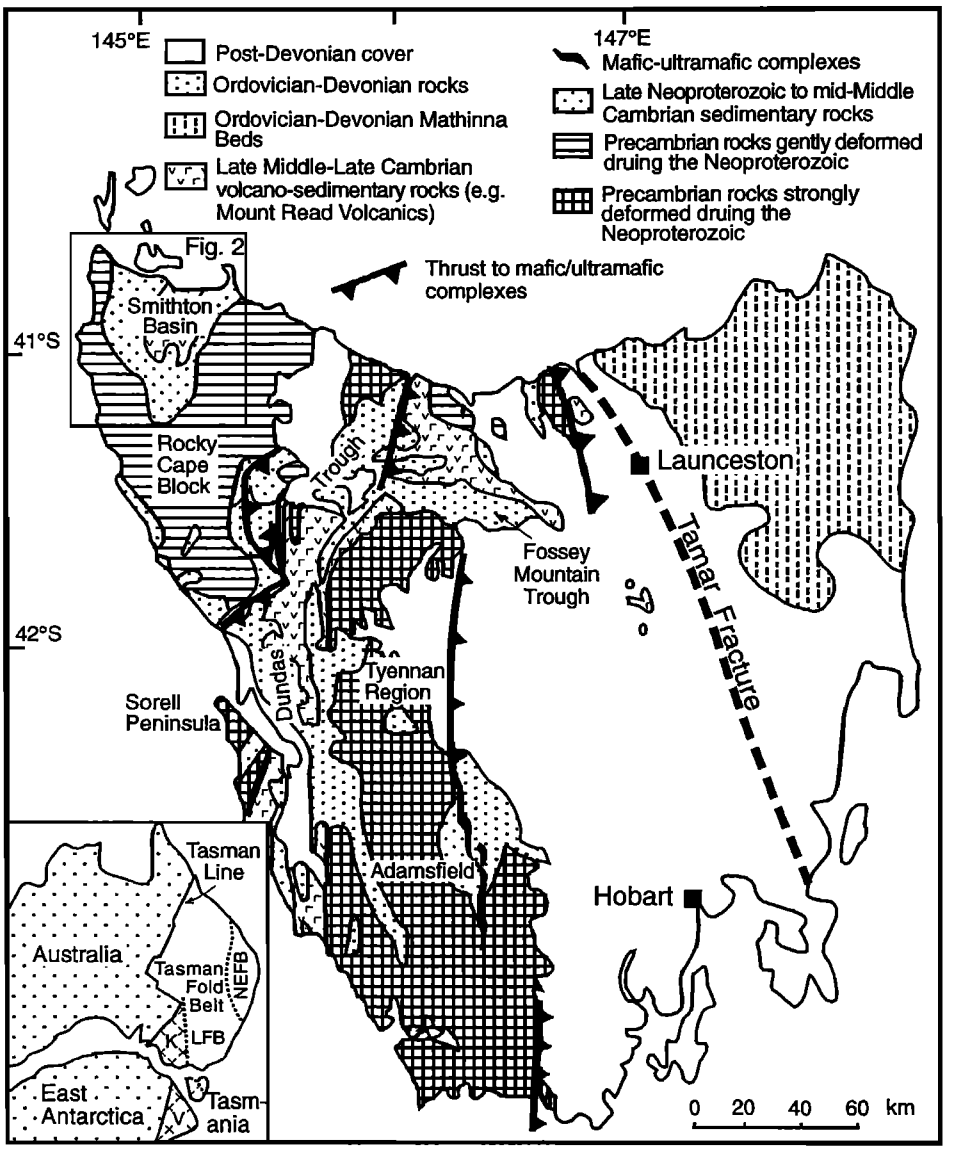


Figure 1. Geological framework of Tasmania [modified after *Crawford and Berry*, 1992]. Inferred position of thrusts shown beneath post-Cambrian cover rocks. LFB, Lochlan Fold Belt; NEFB, New England Fold Belt; K, Kanmantoo Fold Belt; V, Northern Victoria Land.

Neoproterozoic to Upper Cambrian intercalated sedimentary and volcanic rocks resting with slight angular unconformity on folded basement rocks of the Upper Proterozoic Rocky Cape Group [*Lennox et al.*, 1982; *Baillie and Crawford*, 1984] (Figure 2). Basal rocks of the Smithton Basin succession comprise locally developed siliciclastic conglomerate and sandstone (Forest Conglomerate and Quartzite) overlain by stromatolitic-bearing dolomitic carbonate rocks (Black River Dolomite), followed with apparent conformity by a mixed basalt/sedimentary succession (Crimson Creek Formation correlate) probably formed in a continental rift environment [*Baillie and Crawford*, 1984; *Brown*, 1989]. An overlying unfossiliferous carbonate succession (Smithton Dolomite) is in turn overlain by sedimentary rocks containing upper Middle and Upper Cambrian fossils [*Jago and Buckley*, 1971; *Jago*, 1976; *Rickards et al.*, 1990; *Jago and Baillie*, 1992].

The sampled rocks described herein belong to the uppermost unnamed succession of early Late Cambrian (Idamean) age, a correlate of the Dundas Group of western Tasmania, and crop out on the shore platform at Stony Point some 16 km northwest of Smithton in far northwest Tasmania (Figure 2; Hunter 1:100,000 Sheet, AMG grid reference CQ293875). Two lithofacies associations are recognized: a thin-bedded association of fine grained or very fine grained sandstone, siltstone, and mudstone, and a coarse grained association of thick-bedded coarse grained to very coarse grained sandstone, pebbly sandstone, and granule conglomerate. Both lithofacies are considered to have been deposited within a submarine basin floor turbidite-fan complex [*Baillie and Jago*, 1995]. The coarse grained association is interpreted as a succession of channel-fill deposits and the thin-bedded association, from which all of our samples were collected, as overbank deposits consisting predominantly of mud

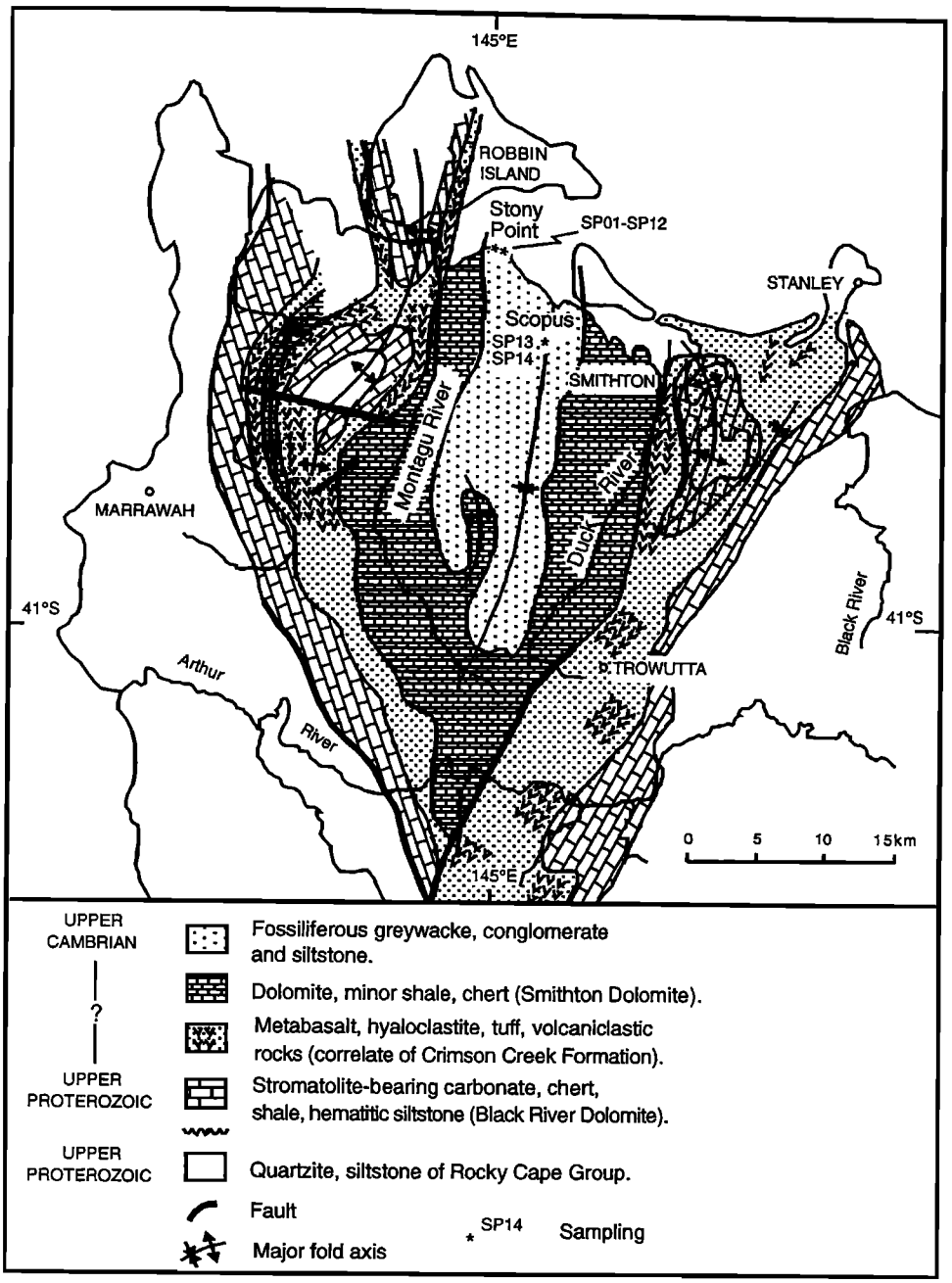


Figure 2. Geology of the Smithton Basin [after *Baillie and Jago,* 1995]. Stars show sampling localities.

turbidites. Petrographic data indicate that the sandstones were derived predominantly from magmatic arc material from both felsic and mafic volcanoes, but that a metamorphic source similar to that currently observed in Tasmanian Precambrian terrains was a supplier of some sediment. Samples were mostly collected from red siltstones/mudstones within the thin-bedded association cropping out around the 150m mark on the logged section of *Baillie and Jago* [1995] (sites SP01-SP12). The rocks from the surrounding shore platform are mildly deformed, uncleaved, and dip east at around 35°–45°. No mesoscopic folds were observed in outcrop and limited structural data suggests that fold axes are subhorizontal. There are no upper age constraints on the age of the folding although from regional geology it is likely to be either Late Cambrian to Early Ordovician (Delamerian) or late Middle Devonian (Tabberabberan). Sites SP13 and SP14 were from dark grey mudstones and turbiditic siltstones at a quarry (Scopus), from where lower Upper Cambrian trilobites and dendroids have been recorded [*Rickards et al.*, 1990].

Paleomagnetic Results

AMS Results

Anisotropy of low-field magnetic susceptibility (AMS) of the samples was measured using a Bartington MS2 meter prior to demagnetization. The purpose was to detect the level of strain in the rocks, which may have affected the remanent directions. A

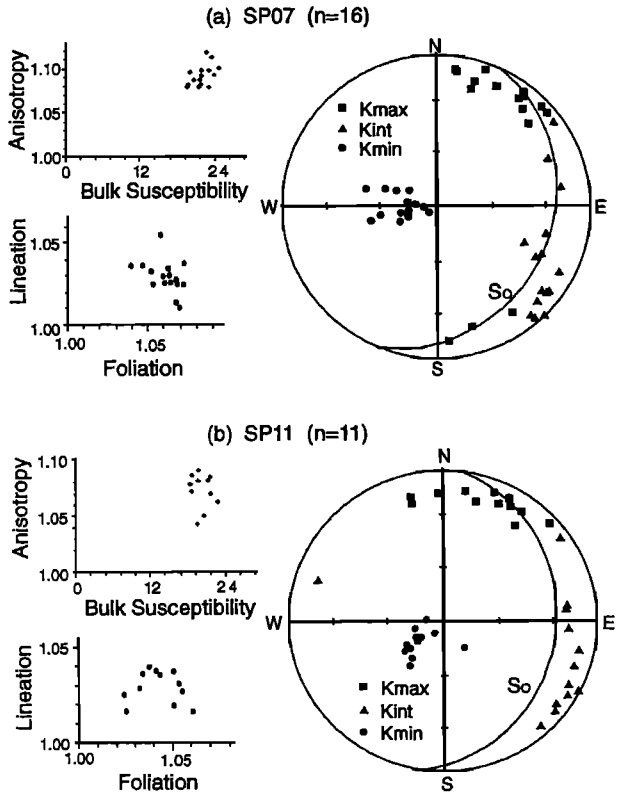


Figure 3. Representative plots of anisotropy of magnetic susceptibility (AMS) results. Each dot represent one specimen, with "n" representing the total number of specimens measured: (a) Figure shows results from site SP07 and (b) results from site SP11. Unit for susceptibility is $1x10^{-6}$ CGS (4π $x10^{-6}$ SI). Anisotropy = K_{max}/K_{min}; Lineation = K_{max}/K_{int}; Foliation = K_{int}/K_{min}. So, bedding attitude.

large majority of the samples has a bulk magnetic susceptibility of ~25 x 10^{-6} CGS (~30 x 10^{-5} SI), and an anisotropy degree (K_{max}/K_{min}) between 1.02 and 1.10. All 14 sampling sites exhibit a magnetic foliation subparallel to the bedding plane, with the majority of the sites having a weakly developed NNE-SSW trending magnetic lineation in both geographic and stratigraphic coordinates (e.g., Figure 3). We interpret the magnetic foliation as the result of compaction. The weak magnetic lineation could either represent the paleocurrent direction, as it is consistent with the dominant northerly trending paleocurrent directions measured in outcrop [*Baillie and Jago*, 1995], or be due to the mild tectonic compression, which could produce a magnetic lineation parallel to the fold axis. In either case, the compaction-dominated magnetic fabric precludes any significant effect of internal strain on the magnetic remanent directions.

Magnetic Remanence Results

One or two standard paleomagnetic specimens from each oriented sample were subjected to demagnetization and remanence measurement procedures. Samples from sites SP01–SP12, which were mostly from red siltstones and mudstones and have natural remanent magnetization (NRM) intensities between 5 and 10 mA/m, were demagnetized using either a Schonstedt TSD-1 furnace or a Molspin 2-axis tumbler alternating field (AF) demagnetizer, and measured using an upgraded Digico spinner magnetometer. Samples from sites SP13 and SP14 were from black mudstones and turbiditic siltstones, with NRM intensities mostly below 1 mA/m. These samples were demagnetized using two MMTD1 thermal demagnetizers manufactured by Magnetic Measurements, and an automated 2G600 AF demagnetizer, and their remanence was measured using a 2G 755-R cryogenic magnetometer (magnetic moment noise < 4 x 10^{-3} mA/m RMS per root Hz). All instruments are hosted in low-field (hundreds of nanoteslas) spaces created by room-size Helmholtz coils.

Orthogonal and stereographic plots were used for identifying linear vectors, and the principal component analysis technique [*Kirschvink*, 1980] was used for calculating the best-fit directions. Remanence components in the samples can generally be clearly separated, with examples shown in Figures 4 and 5. Eleven out of the 14 sites revealed both a "soft" component (unblocking temperature <600°C) and a more stable ("hard") component (Figure 4). The "hard" component revealed two opposite polarities from the two sampling successions at Stony Point. In some samples with the interpreted normal polarity remanent direction, there are signs of a small superimposed reverse component (e.g., Figure 4h). This "hard" component is resistant to AF demagnetization, and displays square demagnetization shoulders in its intensity decay curves when demagnetization temperatures approach 700°C (Figure 4), indicating hematite as its carrier mineral.

In situ directions of the high-unblocking temperature component are shown in stereographic plots in Figure 6a-6k with effect of the bedding-correction on the site-mean directions shown in Figure 6l-6m. All but site SP06 (Figure 6f) have their sample-mean directions well grouped, and directed away from the present geomagnetic field direction (P) and dipole field direction (*D*) (Figure 6l). A correlation test (definition 2) of *McFadden* [1990] on the 11 sites shows that the *in situ* directions correlate with bedding attitude at the 99% confidence level, but the bedding-corrected directions have no apparent correlation with bedding attitude, indicating a prefolding origin. A summary of the mean directions of this prefolding remanence is given in Table 1.

"Soft" components have been identified from all sites. In sites SP12, SP13, and SP14, they are the only clearly defined components, although the demagnetization path of most samples does not converge on the origin (Figure 5), indicating the existence of a more stable component.

Directions of the "soft" components are listed in Table 2, and plotted in Figure 7. *In situ* directions from 11 out of the 14 sites fall in a tight group in the northwest quadrant (component S1), whereas directions from sites SP04, SP13, and SP14 fall in the northeast quadrant (component S2) (Figure 7o). S2 can be easily demagnetized by both thermal and AF demagnetization (e.g., Figure 5d), indicating magnetite as the probable magnetic carrier. However, AF demagnetization is not always effective in removing S1 (Figure 4a and 4d), indicating magnetite, and possibly some very fine-grained hematite, have contributed to S1. S1 failed the correlation test (definition 2) of *McFadden* [1990], with the *in situ* SCOS2 = 3.3, bedding-corrected SCOS2 = 8.0 (the 95% SCOS for 11 sites is 3.9, and 99% SCOS = 5.4), indicating that S1 is a post-folding magnetic overprint. The *in situ* mean direction of D= 304.3°, I = -77.2°, α_{95} = 3.7°, and k = 154.5 for S1 gives a paleo-South Pole position at (50.4°S, 177.4°E), DP = 6.5°, DM = 6.9°. The tightly grouped S2 is also

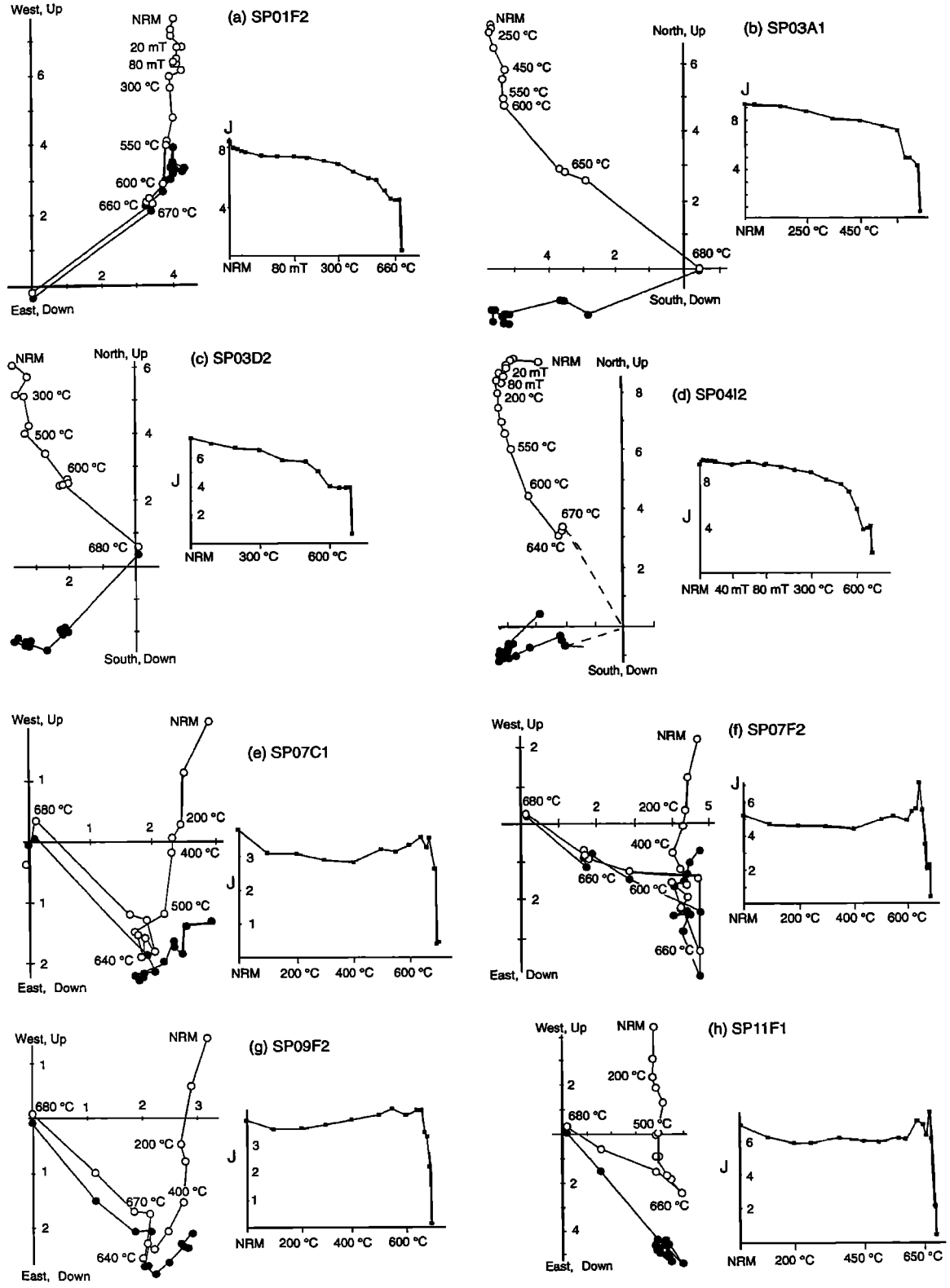


Figure 4. Representative orthogonal plots of demagnetization results (*in situ*) from samples which contain the interpreted primary magnetic remanence. Solid circles are projections on the horizontal plane and open circles are projections on the vertical planes. Unit of magnetic remanence is milliampere/meter.

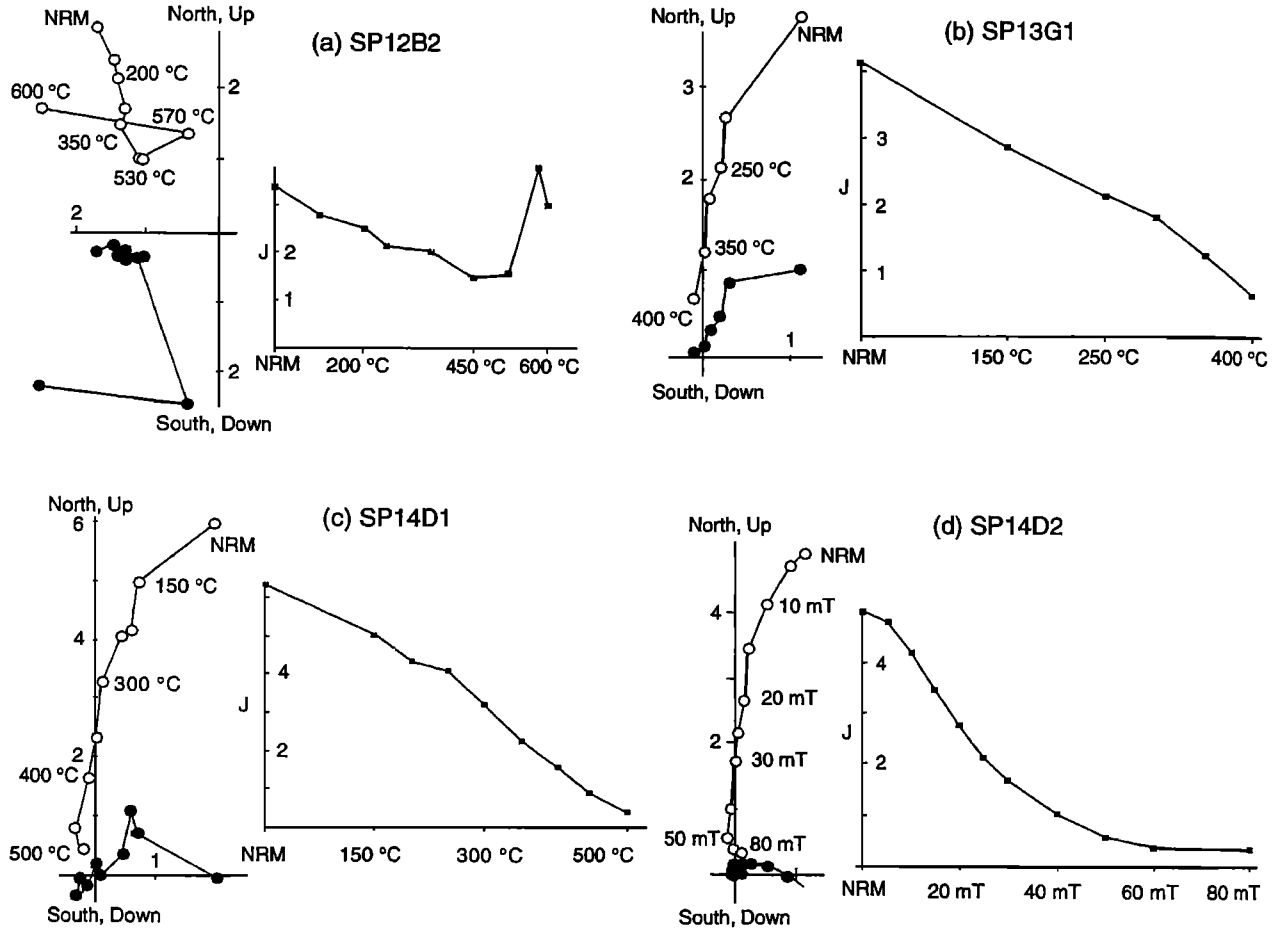


Figure 5. Representative orthogonal plots of demagnetization results from samples in which only magnetic overprints can be defined. Conventions are the same as in Figure 4.

interpreted as an overprint. Its overall mean of D = 35.5°, I = -72.0°, N=3, α_{95} = 6.4°, k = 44.6 gives a South Pole position at (62.0°S, 102.6°E), with DP = 9.9°, DM = 11.3°.

Tectonic Significance

Late Cambrian Primary Remanence: Connection of Northwestern Tasmania with Gondwanaland

Although significant volcanic activity occurred in the Smithton Basin and along the northeasterly trending Dundas "Trough" at the southeastern margin of the Rocky Cape block at various times during the Cambrian [e.g., *Berry and Crawford*, 1988; *Crawford and Berry*, 1992], the mid–Cambrian to Early Ordovician Ross–Delamerian Orogeny seems to have had little effect on rocks in the Smithton Basin [e.g., *Findlay*, 1987]. Major deformation in western Tasmania probably did not occur until the late Middle Devonian Tabberabberan Orogeny, when there was broad folding and thrusting in northwestern Tasmania [e.g., *Leaman et al.*, 1994]. The "hard" magnetic remanent component, which passes the fold test, should therefore be a remanence acquired before the Late Devonian.

Figure 8 presents the existing paleomagnetic poles of Cambrian and Early Ordovician age from East Gondwanaland as listed in Table 2 of *Li and Powell* [1993], all of which satisfy at least the "C" class quality requirements. Additional poles include an "A" class pole from the Cambrian–Ordovician boundary Black Mountain carbonates (pole BM) of the Georgina Basin [*Ripperdan and Kirschvink*, 1992], which falls at (3.1°N, 54.1°E) with DP = 7.4° and DM = 14.8°; a "C" class poles from interpreted Lower Cambrian dolerite intrusions in the Albany Mobile Belt of West Australia (pole AD) [*Harris and Li*, 1995], which falls at (37.8°S, 346.9°E) with A_{95} = 11.7°; a "C" class pole from the Early Cambrian Purple Sandstone (pole PS) of India [*McElhinny*, 1970], which falls at (28°S, 32°E) with A_{95} = 11° in present Indian coordinates; an "A" class pole from the Lower Ordovician igneous rocks in East Antarctica (pole SVL) [*Grunow*, 1995], which falls at (3.5°S, 23.0°E) with A_{95} = 5.9° in present Antarctica coordinates, and a "C" class pole from the Lower Ordovician Black Hill Norite in South Australia (pole BH) [*Schmidt et al.*, 1993], with an averaged pole position at (38.7°S, 26.3°E) with A_{95} = 8.7°. The general trend of apparent polar wandering (APW) for East Gondwanaland during the Cambrian and Early Ordovician, shown in the Gondwanaland configuration after *Powell and Li* [1994], starts from the western margin of South America in the earliest Cambrian, moves across western Africa during the Middle to Late Cambrian, and returns to western Africa by the Early Ordovician after doing a hairpin at the Cambro-Ordovician boundary. For simplicity, poles from

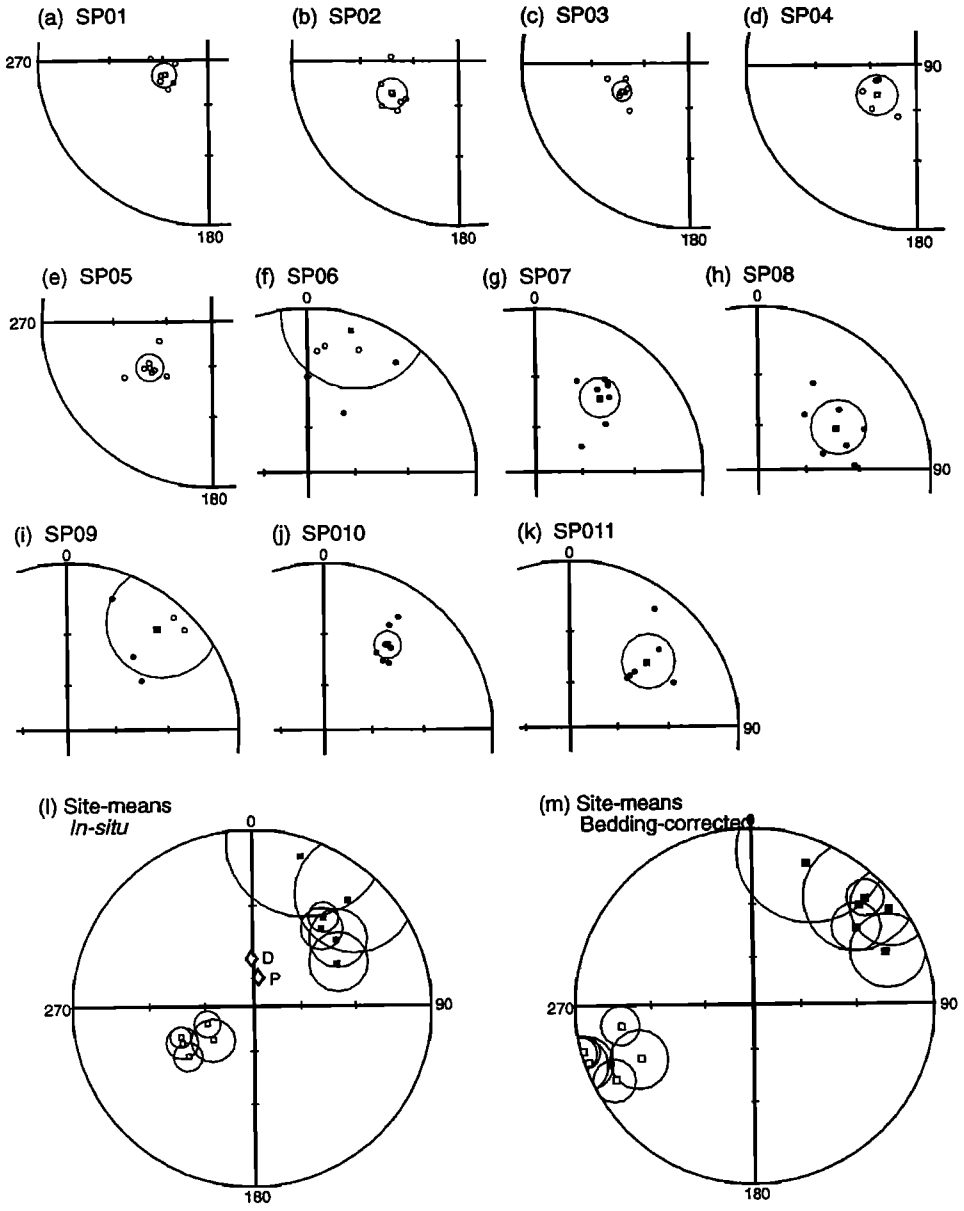


Figure 6. Stereographic plots of the interpreted primary magnetic remanent directions. Dots (circles) represent sample-means, and squares represent site-means. Small circles show the α_{95} of the site-means. Open symbols are upper hemisphere projections, and solid symbols lower hemisphere projections. Figures 6a-6l are *in situ*, and Figure 6m is bedding-corrected. *D*, present dipole field direction; *P*, present geomagnetic field direction. Equal-angle projection.

West Gondwanan cratons are not shown, although their positions merge with those of East Gondwanaland during the Cambrian [*Li and Powell*, 1993; *Powell et al.*, 1993; *Meert et al.*, 1995].

The pole from the lower Upper Cambrian turbidites at Stony Point in northwestern Tasmania obtained by this study (pole SP, Table 1) falls on the corresponding position of the East Gondwanan APW path, although the APW path passes this position again during the Early Ordovician (Figure 8). Such a pole position is distinct from all the known younger paleopole positions for Australia [e.g., *Li et al.*, 1993]. The SP pole is therefore either an early Late Cambrian primary pole, or an Early Ordovician overprint pole.

If the SP pole is primary, it implies that no large-scale transcurrent movement (i.e., movement of more than a few hundred kilometers), or significant rotation, has occurred between the Rocky Cape block of northwestern Tasmania and cratonic Australia since the early Late Cambrian. That is, the Rocky Cape block has been part of East Gondwanaland since at least early Late Cambrian. If Tasmania were rifted from East Gondwanaland during the breakup of Rodinia in the late Neoproterozoic, as suggested by *Powell et al.* [1994], its accretion onto East Gondwanaland should have occurred before the Late Cambrian.

If the SP pole represents an Early Ordovician overprinting

Table 1. Results of the Interpreted Primary Magnetic Remanence
Location (40.7°S, 145.0°E)

Site	Bedding Attitude		*In Situ*		Bedding-Corrected		Common		Polarity
	Dip/Dip-Azimuth, degree	n/no	Dec, degree	Inc, degree	Dec, degree	Inc, degree	α_{95}, degree	*k*	
SP01	45.2/92.5	6/8	250.8	-59.8	261.3	-16.1	7.9	72.7	*R*
SP02	45.2/92.5	7/8	243.5	-42.6	251.6	-1.2	8.9	47.0	*R*
SP03	45.2/92.5	7/8	248.0	-43.2	254.9	-0.7	5.7	113.4	*R*
SP04	40.9/81.7	5/9	231.3	-58.1	245.2	-20.1	12.7	37.4	*R*
SP05	38.5/85.5	7/9	232.9	-41.5	241.5	-7.2	7.8	61.3	*R*
Mean of	reversed polarity:	N = 5			250.9	-9.1	11.1	48.5	*R*
SP06	19.3/117.5	6/9	17.4	6.4	20.1	9.4	26.4	7.4	*N*
SP07	26.8/113.6	8/8	40.6	30.5	52.4	19.8	10.3	29.7	*N*
SP08	24.0/96.4	7/8	61.7	35.3	67.7	14.8	14.7	17.9	*N*
SP09	19.5/131.6	5/8	41.1	13.2	45.6	12.6	22.5	12.5	*N*
SP10	29.6/98.8	7/8	37.4	25.6	45.4	9.3	6.8	80.1	*N*
SP11	25.5/79.2	6/7	50.3	28.7	54.0	5.9	13.7	24.8	*N*
Mean of	normal polarity:	N=6			47.5	12.3	13.5	25.6	*N*
Total mean (N = 11, SCOS 95% = 3.9, 99% = 5.4):			48.4	36.0	58.3	11.1			
			α_{95} = 12.1°		α_{95} = 10.4°				
			k = 15.1		*k* = 20.2				
			SCOS2 = 5.5		SCOS2 = 1.4				

Bedding attitude is averaged from several readings. n/no, number of samples revealing the characteristic remanence/total number of samples; N, number of sites; α_{95}, half-angle of the 95% confidence cone for mean-direction; Dec, declinations; Inc, Inclination; *k*, precision parameter of *Fisher* [1953] distribution for mean directions; Polarity, magnetic polarity, where *N* for normal, and *R* for reversed; *DP/DM*, semi-axes of elliptical error around the pole at a probability of 95%; SCOS, critical value of *McFadden* [1990] correlation test statistics, whereas "2" stands for definition 2. Paleo-South Pole position (SP) calculated from the mean direction of the 11 sites is (19.4°S, 28.9°E), *DP* = 5.3°, *DM* = 10.5°.

Table 2. List of Magnetic Overprint Results
Location (40.7°S, 145.0°E)

Site	n/no	Dec., degree	Inc., degree	α_{95}	*k*	Polarity
			"Soft" component S1:			
SP01	8/8	301.1	-79.2	3.4	272.5	*N*
SP02	8/8	251.2	-80.7	7.2	60.7	*N*
SP03	8/8	306.2	-81.4	6.4	75.1	*N*
SP05	7/9	336.8	-79.2	6.7	81.9	*N*
SP06	9/9	299.1	-77.4	7.2	51.8	*N*
SP07	8/8	304.5	-68.6	7.7	52.1	*N*
SP08	8/8	305.7	-74.4	5.8	90.7	*N*
SP09	6/8	312.9	-68.8	6.0	127.7	*N*
SP10	8/8	322.7	-80.6	6.3	78.1	*N*
SP11	5/7	307.0	-81.5	4.1	346.5	*N*
SP12	7/7	292.7	-71.0	8.0	57.6	*N*
Mean of 11 sites:		304.3	-77.2	3.7	154.5	*N*
			"Soft" component S2:			
SP04	7/9	40.9	-72.3	15.0	17.0	*N*
SP13	12/12	42.1	-74.7	5.3	68.3	*N*
SP14	10/12	26.1	-68.6	4.7	107.0	*N*
Mean of 3 sites:		35.5	-72.0	6.4	373.5	*N*

n/no, number of samples revealing the characteristic remanence/total number of samples; N, number of sites; α_{95}, half-angle of the 95% confidence cone for mean-direction; Dec, declinations; Inc, Inclination; *k*, precision parameter of *Fisher* [1953] distribution for mean directions; Polarity, magnetic polarity, where *N* for normal, and *R* for reversed; *DP/DM*, semi-axes of elliptical error around the pole at a probability of 95%. Paleo-South Pole position (TOP3) calculated from component S1: (50.4°S, 177.4°E), *DP* = 6.5°, *DM* = 6.9°. Paleo-South Pole position (TOP4) calculated from component S2: (62.0°S, 102.6°E), *DP* = 9.9°, *DM* = 11.3°

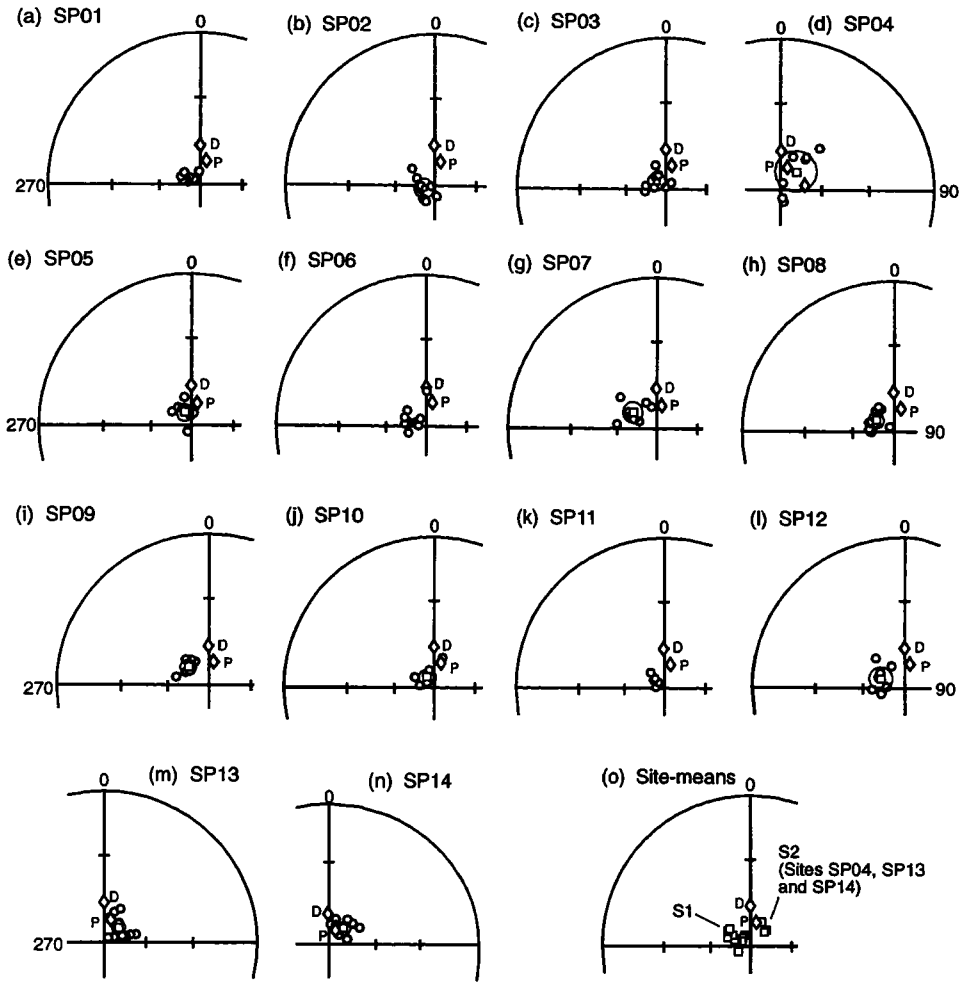


Figure 7. Stereographic plots of the magnetic overprint directions. Other conventions see Figure 6.

event due to fluid circulation during the terminal stages of the Ross–Delamerian deformation, it implies that Tasmania is in its Gondwanaland position adjacent to Australia–Antarctica by the Early Ordovician at the latest, which discounts tectonic models which have the Rocky Cape block in other positions [e.g. *Elliot and Gray*, 1992].

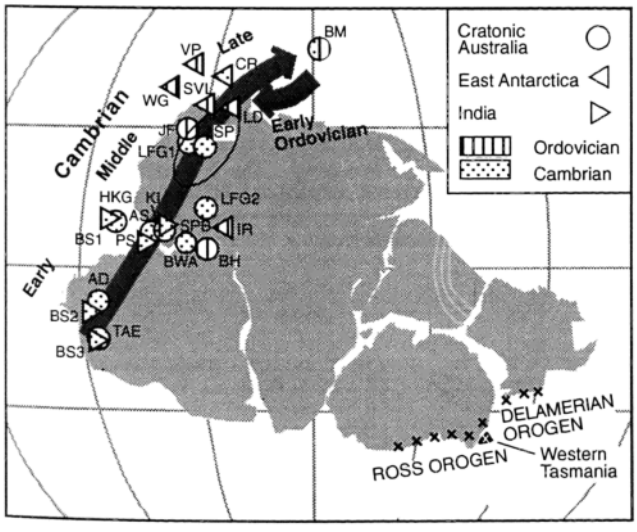


Figure 8. Cambrian and Ordovician paleomagnetic poles from East Gondwanaland as selected by *Li and Powell* [1993], and discussed in text. Gondwanaland, shown in present African coordinates with 30° grids, is constructed using Euler poles given by *Powell and Li* [1994]. Pole SP is the interpreted late-Early Cambrian pole obtained by this study from northwestern Tasmania, which is in agreement with other East Gondwanan results.

Mesozoic–Cenozoic Overprints: Thermal Events During the Breakup of Gondwanaland

The position of the overprint pole from S1 (TOP3) differs from any known late Paleozoic poles from Australia, but is close to the mid-Jurassic (~175 Ma) average pole [*Embleton*, 1984], which was based on results from numerous igneous bodies in southeastern Australia and Tasmania [e.g., *Schmidt and Embleton*, 1981] dated between 190 Ma and 160 Ma (Figure 9). This overprint is therefore likely to be due to a mid-Jurassic thermal/chemical event, probably related to the widespread volcanic activities in southeastern Australia and the development

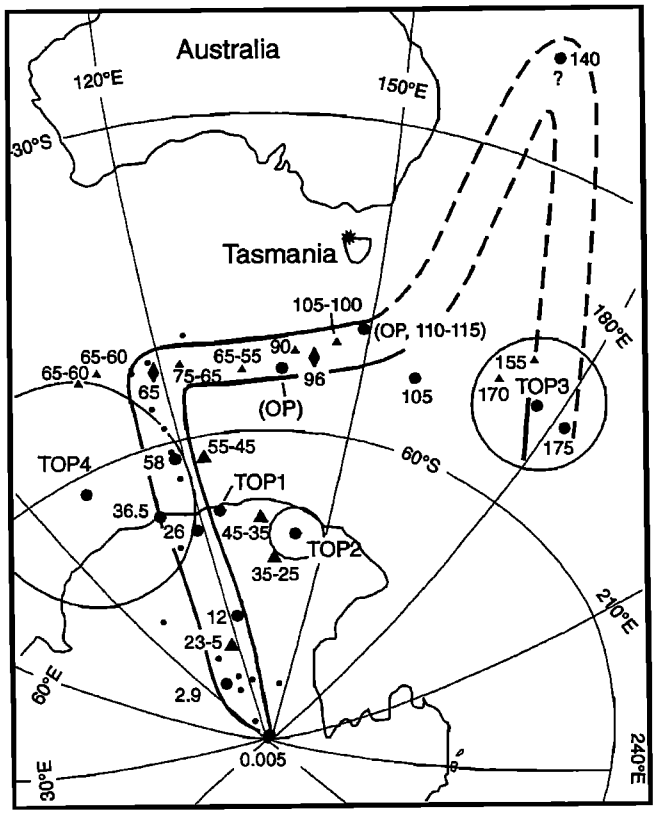


Figure 9. Mid-Mesozoic–Cenozoic APWP of Australia and positions of the overprint poles from Tasmania (TOP poles, see discussion in text). Base map was modified after *Veevers and Li* [1991], with an additional pole dated as 36.5 Ma from Australia [*Idnurm*, 1994].

of the widespread Tasmania dolerites (170.5 ± 8.0 Ma) [*Schmidt and McDougall*, 1977] and the Ferrar Supergroup intrusive/extrusive rocks along the Transantarctic Mountains [179 ± 7 Ma, *Kyle et al.*, 1980]. These mid-Jurassic igneous activities have been interpreted as the precursor of the breakup of Gondwanaland [e.g., *Storey*, 1995].

The pole from component S2 (TOP4) falls close to the early Tertiary part of the APW path (Figure 9). We interpret this component as either an Early Tertiary overprint, or a viscous remanence acquired in the recent geomagnetic field, as suggested by the close alignment of the S2 directions with the present field (Figure 7d, 7m, 7n, and 7o), and by the low unblocking temperature and coercivity of this component.

Post-Paleozoic magnetic overprints have been shown by previous studies to be widespread, and in places pervasive, in Tasmania. *Briden* [1967] reported NRM results of a pilot investigation over Paleozoic rocks in central and northwestern Tasmania (including this study region), and found that all but one sampling locality gave a Tertiary overprint pole position (TOP1) at (68°S, 124°E). Although no routine demagnetization was performed on those samples, the position of pole TOP1 is close to TOP4 of this study. *Sharples and Klootwijk* [1981] reported a pervasive overprint from the Ordovician Gordon limestone in southern Tasmania, with a pole position at (59.8°S, 141.1°E), $DP = 2.4°$, $DM = 2.5°$ (pole TOP2 in Figure 9). The overprint was interpreted as of Late Cretaceous, or less likely, Early Tertiary age, caused by the rifting events before the opening of the Tasman Sea. It is therefore possible that there have been multiple phases of overprints in Tasmania due to the rifting events surrounding it during the mid-Jurassic to Early Tertiary interval, although we do not rule out the possibility that the younger overprints are actually recent viscous magnetizations.

Conclusions

A paleomagnetic study on lower Upper Cambrian turbiditic red siltstone and mudstone in the Rocky Cape block of northwestern Tasmania revealed a possible primary or early diagenetic remanence that predates the late Middle Devonian deformation. The corresponding paleo-pole at (19.4°S, 28.9°E) with $DP = 5.3°$, $DM = 10.5°$ agrees with Late Cambrian and Early Ordovician pole positions from East Gondwanaland. We suggest that northwestern Tasmania either stayed close to East Gondwanaland during the breakup of Rodinia at about 700 Ma, or, if it rifted away from the major cratons, it was accreted back to East Gondwanaland by the Late Cambrian or Early Ordovician. Tectonic models which place Tasmania in a position with respect to Australia other than its Gondwanaland position after the Early Ordovician are therefore discounted. However, paleomagnetic information from rocks older than the Late Cambrian is required to examine the earlier history of the Rocky Cape block and its role during the breakup of Rodinia.

Magnetic overprints from Tasmania suggest that there may have been more than one thermal event in the region during the breakup of Gondwanaland. However, primary remanence may still be retained in rocks resistant to thermal overprint (such as red siltstone and mudstone), and in relatively tectonically stable regions like the Rocky Cape block.

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